

VLM Fundamentals & Transformers

By
Niloy Farhan
Lecturer
Department of CSE,
SDS, BRACU

CVIS Hands-on Training on Development of Machine
Learning and Deep Learning Models Summer 2025



Inspiring Excellence



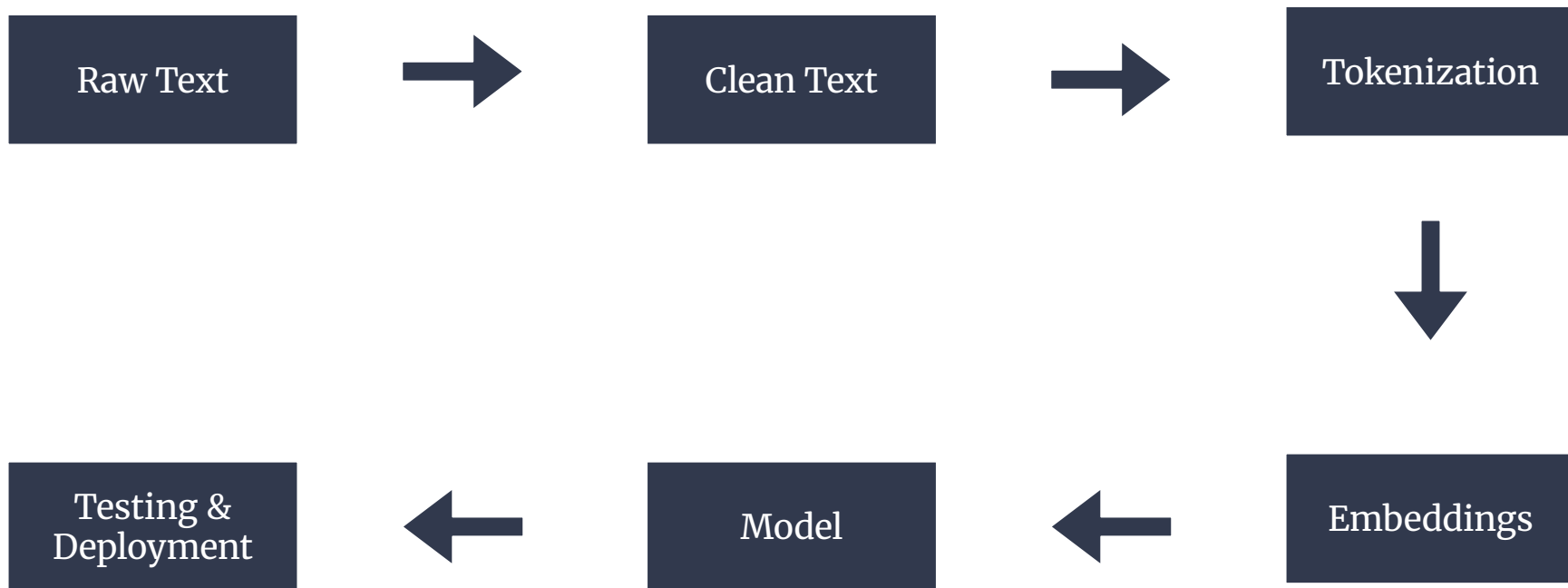
Overview

- The NLP Pipeline: From Raw Text to Data
- Hands-on: Text Cleaning & Tokenization
- Word Embeddings: Giving Words Meaning
- Hands-on: Using GloVe & Building an LSTM
- The Transformer Revolution

Why Can't We Just Feed Text to a Model?

- Humans understand language with context
- Machines understand numbers
- **The Challenge:** How do we convert unstructured text into a structured, numerical format a model can learn from?
- The Answer: The NLP Pipeline.

The NLP Pipeline



Why Text Preprocessing Matters

- **Raw Text is Noisy & Unstructured:**

Human language contains inconsistencies like capitalization, punctuation, slang, and typos that machines cannot inherently understand.

- **The Core Goal:**

To transform messy text into a clean, structured set of tokens (meaningful units), forming the foundation for all text-based models.

Common Preprocessing

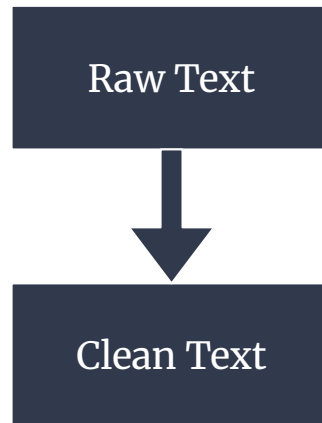
Lowercasing

Punctuation removal

Stopword removal

Stemming vs Lemmatization

Library used: RegEx, NLTK, spaCy



Example:

Raw Text: OMG!!! This movie was ABSOLUTELY fantastic!! 🌟 Best. Film. Ever. #MustWatch

CleanText: omg movie absolutely fantastic best film ever mustwatch

Tokenization

Splitting text into smaller units (tokens)

Tokens can be words, subwords, or characters

Word-level vs Subword-level (BPE, WordPiece)



Example:

omg movie absolutely fantastic best film ever mustwatch

['omg', 'movie', 'absolute', '###ly', 'fantastic', 'best', 'film', 'ever', 'must', '###watch']

How do we represent word into a vector?

The Old Way: Sparse, high-dimensional vectors

One Hot Encoding

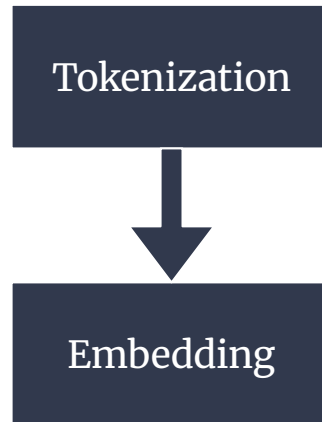
Bag of Words (BoW)

TF-IDF

The New Way: Dense Word Embeddings (Word2Vec, GloVe)

Dense, lower-dimensional vectors (300 numbers)

Capture semantic meaning and relationships



A Good Embedding Should Have

Semantic and Syntactic Relationships:

it should capture meaningful relationships between words

Contextual Similarity:

Words that appear in similar contexts throughout the training data should have similar vector representations, reflecting their interchangeable use

Ability to Solve Analogies:

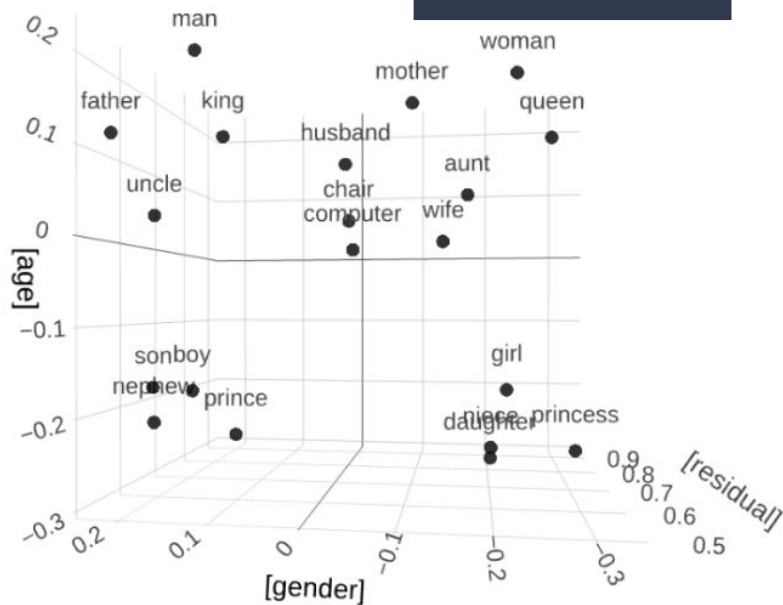
Example:

$$\text{vec}(\text{King}) - \text{vec}(\text{Man}) + \text{vec}(\text{Woman}) \approx \text{vec}(\text{Queen})$$

Tokenization



Embedding



Word2Vec

Neural network model introduced by Mikolov et al. in 2013

Represents words as dense vectors in a continuous vector space

Captures semantic and syntactic relationships between words

Two Approaches

CBOW (Continuous Bag of Words):

Predicts a word from its surrounding context words.

Skip-gram:

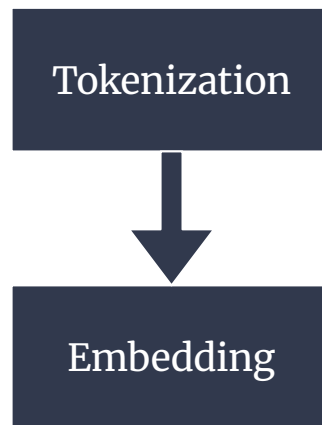
Predicts surrounding context words given a target word.

Example:

king $\rightarrow$ [0.21, -0.17, 0.55, ...]

queen $\rightarrow$ [0.20, -0.19, 0.56, ...]

Here, vector distance shows semantic similarity.

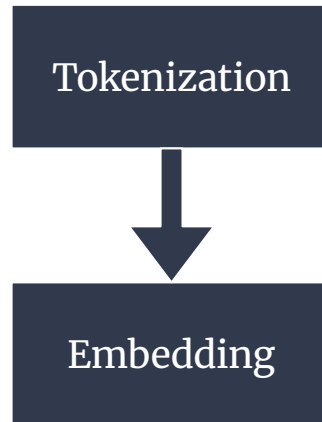


GloVe: Global Vectors for Word Representation

Word embedding model from Stanford (Pennington et al., 2014).

Learns dense word vectors using global co-occurrence statistics of words in a corpus.

Combines the strengths of matrix factorization and Word2Vec-style local context prediction.



Example:

king $\rightarrow$ man $\rightarrow$ [0.52, -0.13, 0.44, ...]

queen $\rightarrow$ [0.54, -0.11, 0.47, ...]

Difference between man and woman aligns with gender dimension.

Limitations of Word2Vec & GloVe

Static Embeddings

Each word has only one fixed vector

Cannot handle polysemy (same word, multiple meanings)

Example: “bank” → same vector for river bank and money bank

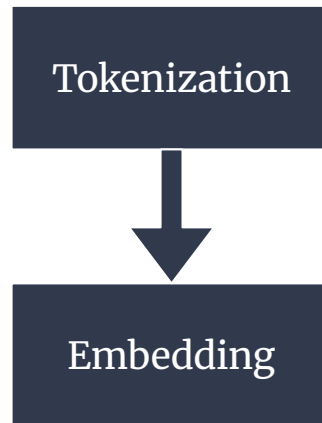
Context Ignorance

Word meaning does not change with surrounding words

Loses nuance in long or complex sentences

Solution?

Dynamic / Contextual Embedding



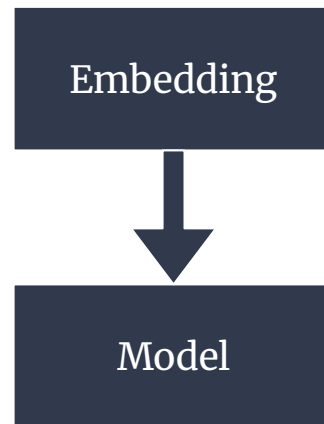
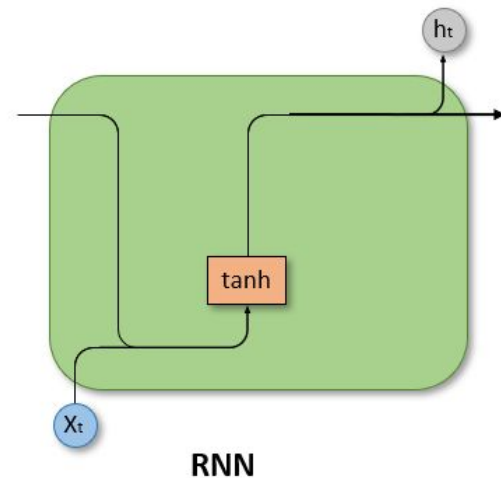
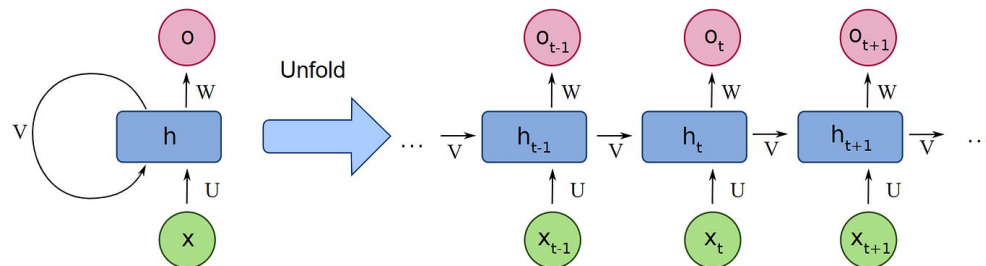
Basic Deep Learning Models for NLP

Recurrent Neural Network (RNN)

Processes sequences step by step

Hidden state passes information through time.

Limitation: Struggles with long-term dependencies
(vanishing gradients).



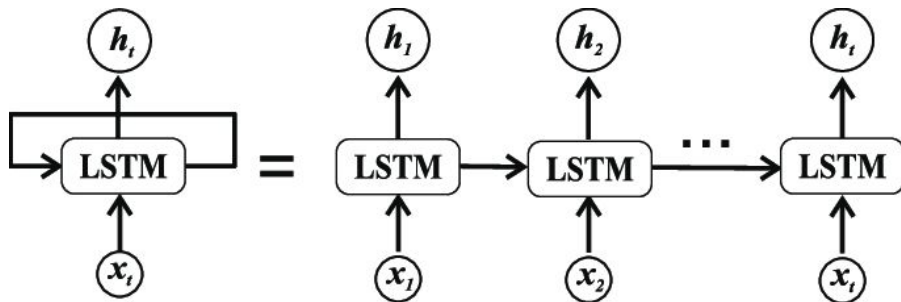
Basic Deep Learning Models for NLP

Long Short-Term Memory (LSTM)

An enhanced RNN with memory cells and gates (input, forget, output)

Retains long-term dependencies effectively

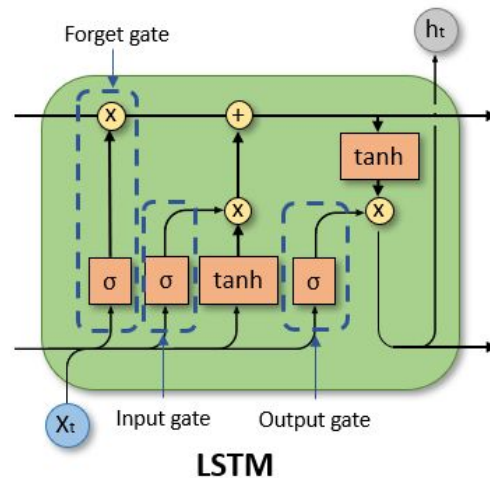
Widely used for text, speech, translation



Embedding



Model



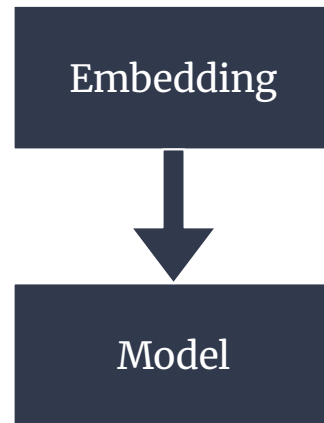
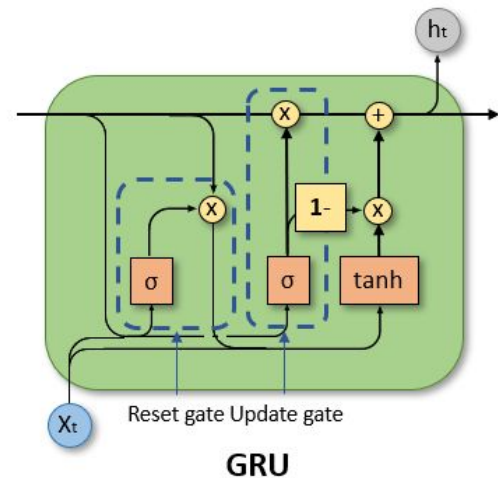
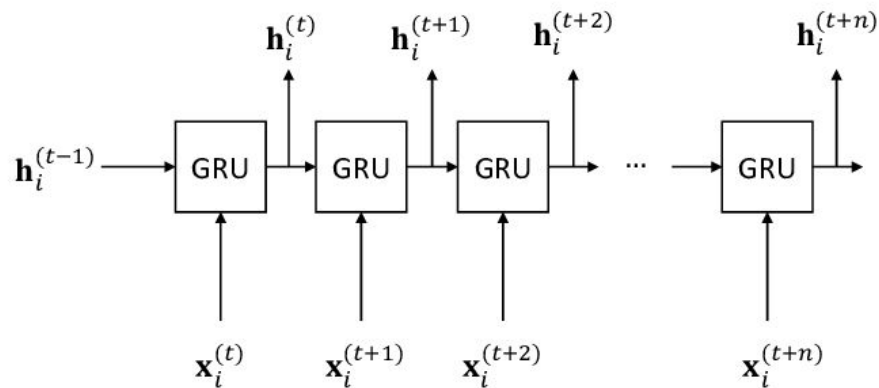
Basic Deep Learning Models for NLP

Gated Recurrent Unit (GRU)

Simplified version of LSTM (fewer gates: reset & update)

Faster to train, performs similarly to LSTM

Popular for NLP tasks with limited resources



Transformers

A Transformer is a type of deep learning model

Introduced by Vaswani et al. in the paper “Attention is All You Need.” in 2017

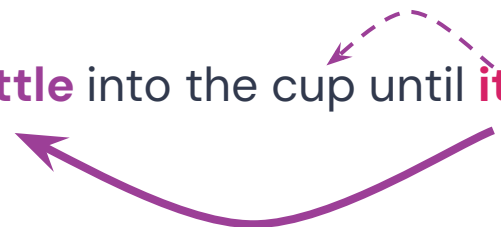
Understands the meaning of words by looking at the context

Powered by Self Attention: Allows the model to focus on important words

I poured water from the bottle into the **cup** until **it** was **full**.



I poured water from the **bottle** into the cup until **it** was **empty**.

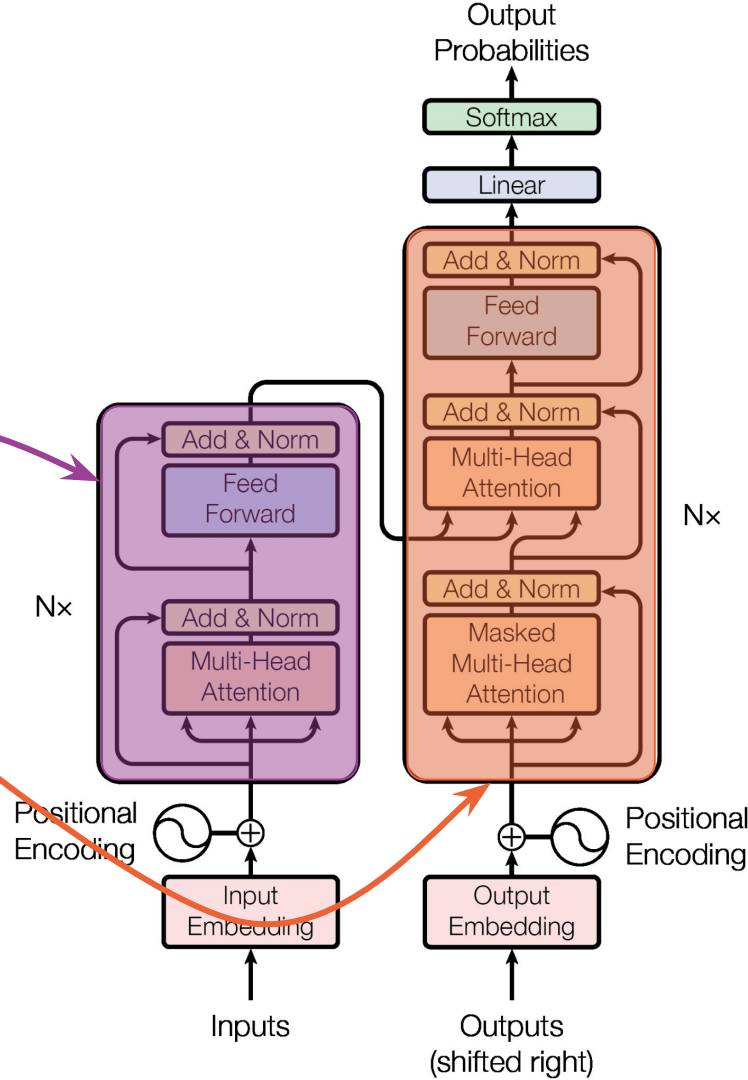


Transformers

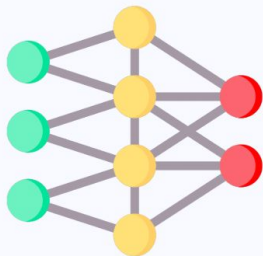
Encoder Decoder Architecture

Encoder: Understands the input

Decoder: Generates the output



LARGE LANGUAGE MODEL - LLM



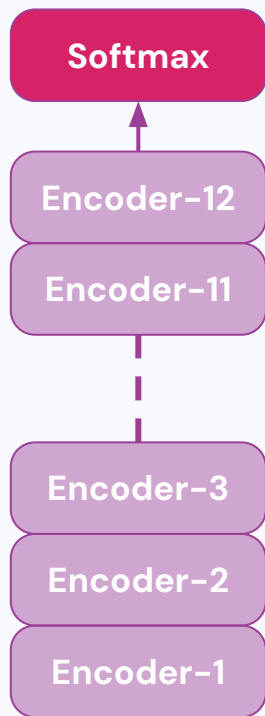
**Transformer
Based
Model**



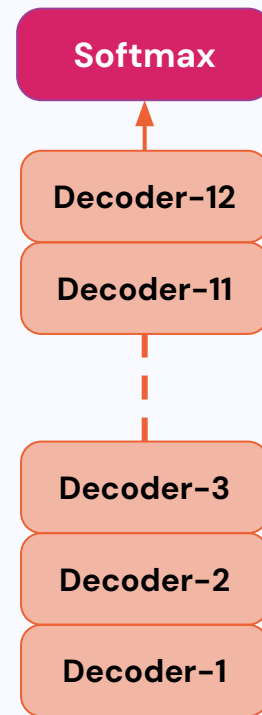
**Trained on
Massive Text
Datasets**



**Autoregressive
Prediction
(GPT-style
models)**



BERT-BASE



GPT 2-SMALL

BERT and GPT

BERT	GPT
Bidirectional Encoder Representations from Transformers	Generative Pre-trained Transformer
Uses only the encoder part of the transformer	Uses only the decoder part of the transformer
12 encoder layers in BERT-base (24 in BERT-large)	12 decoder layers in GPT-2 base (96+ in GPT-4)
Learns context from both directions	Learns context from left to right
Good for understanding tasks (e.g., sentiment analysis, QA)	Good for text generation (e.g., chat, story writing, summarization)

Q&A / Wrap-up

Key Takeaways:

Text must be cleaned and tokenized.

Word embeddings capture semantic meaning in vectors.

Models like LSTMs can process sequences of these vectors.

Transformers use self-attention for context-aware understanding.

Questions?